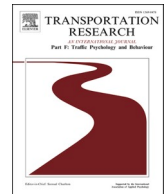




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Understanding drivers' situation awareness in highly automated driving using SAGAT, SART, and eye-tracking data

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ABSTRACT

Rapid and accurately forming situation awareness (SA) is essential when transitioning from autonomous to manual driving. This study examines how drivers' SA is developed when a takeover request (TOR) is issued and compares SA levels across different environmental and time conditions. A laboratory experiment with 39 participants was performed to analyze the SA, perceived safety, and gaze behavior with different levels of traffic density (high and low), road type (urban and highway), and time budget factor (3 s, 10 s, free). The free time budget factor was determined by assessing the participants' response times indicating when they perceived having acquired sufficient SA. The results revealed significant effects of traffic density, road type, and time budget on the Situation Awareness Global Assessment Technique (SAGAT), Situational Awareness Rating Technique (SART), and perceived safety. Post-hoc results for time budget factors revealed significant differences based on the SAGAT score, with the 10-second condition exhibiting the highest score. For the SART and perceived safety scores, the 3-second condition was significantly lower whereas no significant difference was observed between the free and 10-second conditions and the mean response time for the free condition ranged from 10–13 s, with the longest duration in high-traffic and urban conditions. Participants gazed at different AOI when provided with a longer lead time whereas in short lead time conditions, their gaze primarily focused on the front window. These results suggest that drivers develop SA differently based on time constraints, and environmental factors. This study provides valuable insights for developing and implementing autonomous systems, contributing to safer and more efficient transitions between automated and manual driving modes.

1. Introduction

According to the Society of Automotive Engineers (SAE) J3016, vehicle automation is divided into six levels, from 0, indicating no driving automation, to level 5, indicating full driving automation (Society of Automotive Engineers, 2016). Level 3 is defined as conditional driving automation wherein drivers are not required to perform any driving tasks during automated driving, intervening only when prompted by the system to assume control. In this conditionally automated setting, reduced SA is often attributed to drivers performing non-driving-related tasks (NDRTs) during autonomous mode (Liang et al., 2021; Amre & Sun, 2021; Lee et al., 2021; Yoon

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et al., 2019). This decline in SA is primarily influenced by the drivers' disengagement from the driving task owing to reliance on the automated system (Endsley, 2020). Consequently, previous studies have demonstrated that NDRTs impose varying mental and physical demands on drivers which consequently decreases SA which potentially influences their performance during takeover (TO) (Yang et al., 2018).

Although automated systems assist in mitigating human error failures, the roles of human drivers remain crucial. While the system requires the intervention of human drivers at any moment, during a system failure, drivers' situation awareness (SA) remains critical (Endsley, 2020). To reengage control of the vehicle after autonomous driving rapid and precise construction of SA is essential which is affected by the complexity of the situation, density of information, and lead time given. Therefore, this study focuses on how drivers' SA is developed when a takeover request (TOR) is issued and compares the different levels of SA in different environmental and time scenarios. In addition, drivers' gaze behavior to gain SA during TO is investigated to understand the SA acquisition process.

1.1. Situation awareness in highly automated driving

Endsley (1988) defined SA as “the perception of the environment in the current situation, the comprehension of the current situation, and the projection of future status.” SA is crucial during driving because drivers must constantly gather information from the environment to understand the current situation and make decisions in response to rapidly changing events (Gugerty, 2011; Endsley, 2020). In addition to its role in conventional driving, SA is crucial in HAD. In HAD conditions, such as level 3 automation, the vehicle primarily controls the driving task, and drivers intervene only when necessary. These conditions do not demand constant interaction but only partial monitoring of the vehicle, which can negatively impact drivers' SA (Endsley, 2018). Therefore, a decreased level of SA during automation can negatively affect drivers' TO performance, where drivers must reestablish SA within a limited time.

Previous studies have examined drivers' conditions during the automated mode to investigate SA (Röckel & Hecht, 2023). Reduced SA is often attributed to drivers performing NDRTs during autonomous mode (Liang et al., 2021; Amre & Sun, 2021). This decline in SA is primarily influenced by the drivers' disengagement from the driving task owing to reliance on the automated system (Endsley, 2020). Consequently, previous studies have demonstrated that decreased SA affects drivers' ability to regain vehicle control (Yang et al., 2018). This is critical when a TOR is issued in emergencies with a limited time budget, requiring drivers to regain SA within a limited time to reengage in the driving task safely.

Furthermore, research on SA focuses on the driver's state during automation, exploring approaches for developing drivers' SA. Ma et al. (2021) suggested that a two-stage warning system assists in developing a higher SA level than a single-stage system, positively affecting TO performance. Marti et al. (2022) indicated that a two-second interval between the TOR for disengagement from NDRTs and preparation for a TO positively influenced SA compared with being fully involved in NDRTs until the TOR. White et al. (2019) investigated human-machine interface (HMI) strategies to help develop and maintain SA in level 3 conditional automation.

SA in the driving context can be measured in various ways (Salmon et al., 2012). One prevalent measure is the Situation Awareness Global Assessment Technique (SAGAT; Endsley, 1988). Originally designed for aviation fields to assess the SA levels of pilots, the SAGAT measures three different levels of SA. Level 1 SA involves the perception of elements; in the driving context, it is related to information such as vehicle location and color. Level 2 SA involves the comprehension of meaning, which pertains to driving behaviors such as acceleration, braking, and turning. Level 3 SA involves the projection of future states. The SAGAT utilizes the freeze-probe recall technique in a specific situation; thus, it is considered a method for measuring the objective level of SA. This assessment technique is widely used in the automotive domain to measure the level of SA, especially during the takeover task (Ding et al., 2024; Kim et al., 2021; Yang et al., 2023; Zhu et al., 2024). Another measure of SA is the Situation Awareness Rating Technique (SART; Taylor, 2017). This measure is prevalent in other domains and commonly used in the driving context (Tanshi & Söffker, 2022; Van Den Beukel & Van Der Voort, 2013; Tan & Zhang, 2024). The SART is a post-trial subjective rating technique that measures an individual's SA with three subdimensions: attentional demands, attentional supply, and understanding of the situation.

Previous research emphasized the importance of SA for drivers during the automated mode, which affects TO performance. However, knowledge of the required SA levels and their development during TO remains limited. Therefore, this study focuses on the SA level during the transition of control and its impact on readiness and perceived safety.

1.2. Influence of time and environmental factors on situation awareness

Building on previous literature, internal and external factors during TO events influence the reconstruction of drivers' SA. Internal factors encompass aspects such as the lead time available to the driver before the resumption of manual driving and the nature of tasks performed by the driver before the TO request, including NDRTs. External factors pertain to the characteristics of the road on which the automated vehicle is operating and the traffic density around the automated vehicles.

A longer lead time given to drivers to gain SA positively impacts their SA level during the transition of vehicle control (Wu et al., 2023). Tan & Zhang (2021) suggested that drivers' SA is highly dependent on the TO lead time in a noncritical situation with a longer lead time, helping them better understand the road situation. In their subsequent research, they investigated the effect of TOR lead time on drivers' SA in a freeway exit scenario by comparing 12 different lead times in the range of 16–30 s (Tan & Zhang, 2022). The results revealed a positive effect of a longer lead time on the SA level, with a strong improvement during the 8–16 s lead time, reaching a stable level after 20 s. The drivers' decisions to assume control of the vehicle were mostly made at the end of the lead time for all scenarios; that is, they took advantage of the given lead time to develop SA. Zhou et al. (2021) considered the lead time given to participants to gain information to be the most important variable for predicting SA. In addition, the difficulty of the driving scenario significantly affected the SA. The decision time was correlated with SA; thus, a shorter decision time corresponded to an enhanced SA.

Samuel et al. (2016) demonstrated that when participants were provided with over 8 s of lead time to regain vehicle control, the proportion of hazards detected was comparable to that in the control condition in which they were manually driving the vehicle. Similarly, Lu et al. (2017) found that the longer the video length, the lower the absolute error in the number of placed cars, and the lower the total distance error between the placed and actual cars. Notably, these effects appeared to reach saturation at video lengths of 7–12 s. Several studies have found that a longer lead time enhances SA (Lu et al., 2017; Samuel et al. 2016; Wu et al., 2023; Zhou et al., 2021). However, such a correlation may not be applicable when influenced by TO tasks. Some research findings have discussed the influence of the TO task itself on the time required by drivers to achieve SA. That is, drivers tend to resume vehicle control faster despite their lack of sufficient SA (Tan & Zhang, 2022) because they consider the physical reengagement of vehicle control to be more important.

In the environmental context, previous research found that a combination of road conditions, emergencies, and traffic density factors determines the complexity of driving situations. Jia & Du (2023) studied the influences of traffic density, vehicles in the road environment, and lanes on TO performance and SA. Their results revealed a significant main effect for all variables; the more complex the situation, the lower the SA of the participants. The findings of Kim et al. (2022) are consistent with this notion. Li et al. (2023) investigated the correlation between the road environment and SA. The results demonstrated that road curvature affects drivers' SA, influencing their TO performance. The SA recovery time decreased with the curve radius.

1.3. Gaze behavior during the takeover for redeveloping situation awareness

Visual perception is crucial for developing SA in HAD because visual inputs typically contribute to over 70 % of the information individuals gather about the external environment. The eye-tracking measure allows for the measurement of the L1 SA, which is linked to the perception of the environment and is the start of the process. The literature has shown that drivers' visual engagement in NDRTs affects their SA and TO performance (Liang et al., 2021; Yoon & Ji, 2019; Zhou et al., 2021). Previous studies revealed that drivers involved in visually demanding NDRTs had a lower SA or took longer to redevelop SA. Liang et al. (2021) investigated the relationship between drivers' pre-TO visual engagement and their SA level during automated driving. They found that a conscious focus on eye movements, such as fixation on the road, was directly linked to a high SA score. Generally, a longer time spent viewing the driving scene and more dispersed visual attention resulted in better SA. Zhou et al. (2021) developed a predictive model for drivers' SA during the transition period from automated to manual driving. Their model showed that the model with 14 predictor variables outperformed the model with only nine eye-tracking-related variables, such as fixations on the back mirror and road; however, the latter model still predicted drivers' SA at an acceptable level, indicating that drivers' SA could be measured in real-time using only eye-tracking data.

Studies have shown that drivers may look at different areas of interest over time from the TOR. Lu et al. (2017) presented gaze distribution results across four areas of interest: the road center and rear, and the left-wing and right-wing mirrors. The first time the crosshair disappeared, which represents the beginning of the TO, approximately 80 % of the drivers initially focused on the road center. Subsequently, visual attention to mirrors increased, particularly the rear-view mirror. Over time, the proportion of gaze on mirrors decreased, and the attention to the road center increased. Gonçalves et al. (2022) demonstrated that without HMI information, nearly half of the participants gazed at the road center upon disengaging from the NDRTs. However, when the HMI provided information about the post-TO task, such as lane change, the proportion of gaze on the road center decreased, and the proportion of gaze on the HMI was higher than that without HMI information. Acknowledging the significance of gaze behavior in the redevelopment of SA in HAD environments, this study used eye-tracking measures to examine variations in drivers' gaze behavior in response to internal and external situational factors.

1.4. Research objective

Previous research on SA during autonomous driving has primarily focused on the impact of lead time on TO and SA. These studies consistently demonstrate that longer lead times lead to higher levels of SA. Additionally, TO scenarios have been shown to significantly affect SA. However, it remains unclear whether drivers can achieve an SA level comparable to that established with a designated lead time when given sufficient time to assume vehicle control. While some studies have explored driver takeover response times with adequate time buffers (e.g., Tan & Zhang, 2022), there is a lack of research directly comparing SA in predefined takeover situations with that in free-time budget scenarios. Therefore, there is a gap in the research on whether external factors continue to influence SA even in scenarios with sufficient time for TO. To address this gap, this study poses the following research questions:

- RQ1: How do environmental factors (traffic density and road type) and time budget factors (3 s, 10 s, and free) influence drivers' SA (SAGAT and SART) and perceived safety?
- RQ2: How does the response time to indicate sufficient SA differ based on traffic density and road type?
- RQ3: How do environmental and time budget factors affect drivers' gaze behavior when redeveloping SA?

Accordingly, this study aims to determine how different factors of the road environment affect SA and analyze drivers' gaze behavior to develop SA. Therefore, we first analyzed the effects of road type, traffic density, and time on SA and safety. Highways and urban roads were considered as road types with low and high visual distractions, respectively. Time constraint factors were divided into lead time (short: 3 s and long: 10 s) and free response time. The lead time for 3 s and 10 s were set because drivers' situation awareness consistently increases and error decreases until around 7–12 s depending on the contexts (Vlakveld et al., 2018; Lu et al., 2017; Samuel et al., 2016). The major range for time budget from existing studies is located between 3 and 10 s (Deng et al., 2024). This

study primarily focused on how time constraints influence SA levels during the reassumption of vehicle control and analyzed drivers' gaze behavior to develop SA based on eye-tracking data. Moreover, this study investigated how SA is redeveloped during the TO process and the SA level that drivers consider sufficient during the TO task.

2. Methods

2.1. Participants

A total of 39 participants (19 women and 20 men) aged 21–56 years ($M = 36.2$, $SD = 11.19$) were recruited for this study (Table 1). All participants were required to have a valid driver's license and at least one year of driving experience. Most of the participants had significant driving experience and were frequent drivers. A glass-type eye tracker was used, and participants with normal or corrected-to-normal vision with contact lenses were allowed to participate in the experiment. Each participant was paid approximately US\$ 20 as compensation. This study was approved by the Institutional Review Board (IRB) of the Seoul National University of Science and Technology.

2.2. Experiment design and experiment task

The experiment comprised two sessions. The primary objectives of both sessions were to collect data on drivers' SA (SAGAT and SART), perceived safety, and eye behavior. As the time budget factor required different experimental tasks for participants, the experiment was divided into two session.

The first session was designed as a $2 \times 2 \times 2$ within-subject design. Two traffic density levels (high and low), two road types (highway and urban), and two lead-time lengths (3 and 10 s) were considered. Traffic density was controlled by the number of vehicles in the vicinity of the car in which the participants were riding (Fig. 1). The low- and high-traffic-density conditions involved a maximum of two and a minimum of four vehicles near the participant's vehicle, respectively. Two road types were selected in this study: highways and urban roads. The highway had few visual distractors, whereas urban roads were constructed with several visual distractors such as traffic signs, traffic lights, and advertising signs. The lead time for TO was controlled by the length of the video. All conditions were randomly provided during the experiment. The second session aimed to gather the time taken by participants to gain sufficient SA. The experimental design was a 2×2 within-subject framework (road type and traffic density), excluding the lead time. The stimulus control for traffic density and road type was equal to the first experiment. Two repetitions were performed for each condition, yielding eight trials per participant.

In this study, to control the SA level during the automated mode, no road condition information was provided, and participants were asked to be fully engaged in an NDRT. HAD conditions require the driver to assume control of the vehicle after engaging in various NDRTs. This engagement in NDRT might indicate drivers' low SA level or no SA during the automated mode. This implies that the driver has little or no knowledge or information about the driving road situation. This was designed to overcome the limitations of an experimental setting in which the participants' awareness of being tested on TO might influence the SA level during the automated mode.

2.3. Procedure

Before the experiment, the participants were welcomed and asked to sign a consent form indicating the study objectives, and only those who voluntarily agreed to participate in the experiment were included. First, the participants were instructed to fill out a demographic questionnaire that included questions about their age, gender, driving experience, and driving frequency. The experiment was then explained to the participants: introductory material was provided as a written handout, with a complementary video explaining automated driving and TO situations. Subsequently, a practice session on the experimental task was performed to familiarize the participants with the procedure and guide them in acquiring the necessary information to respond to the questionnaires. The participants underwent two practice sessions before each session. The participants were asked to wear an eye tracker before the

Table 1
Participants' Demographic Information.

		n	Proportion
Gender	Male	20	51.3 %
	Female	19	48.7 %
Age	Mean (SD)	36.2 (11.19)	21–56
Driving experience	1–3 yr	5	12.8 %
	3–5 yr	15	38.5 %
	5–10 yr	4	10.3 %
	10 + yr	15	38.5 %
Driving frequency	0–1 per week	3	7.7 %
	2–3 per week	15	38.5 %
	4–5 per week	9	23.1 %
	6–7 per week	12	30.8 %

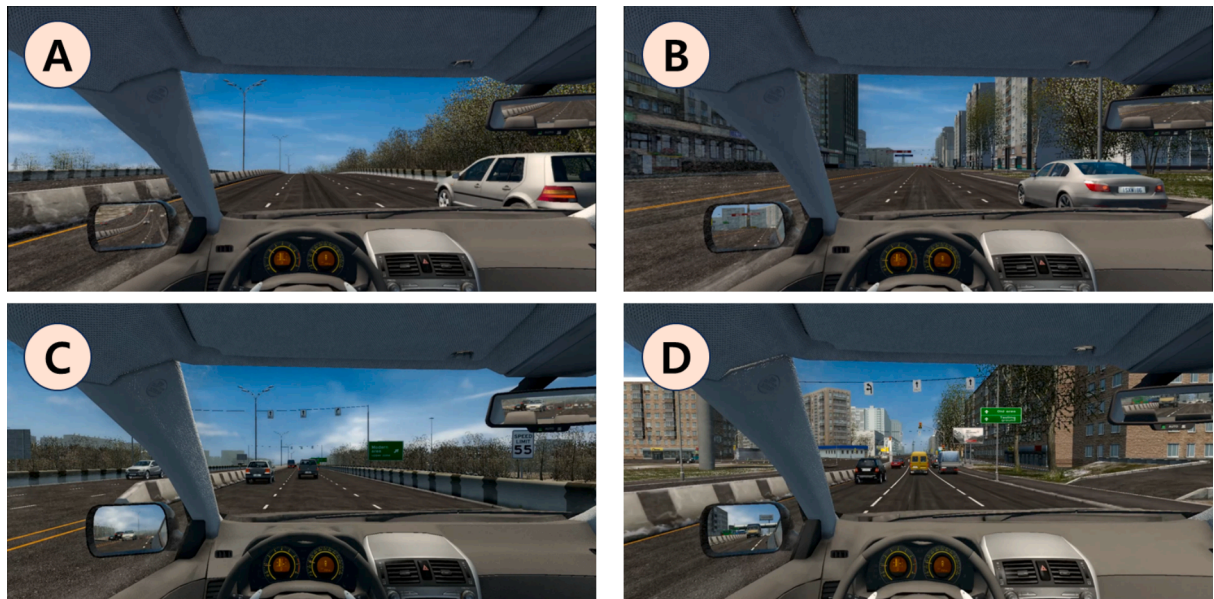


Fig. 1. Snapshot of the experimental conditions. (A) Low traffic density on the highway. (B) Low traffic density in urban areas. (C) High traffic density on the highway. (D) Low traffic density in urban areas.

experiment.

The first session aimed to assess the SA level gained during a predefined lead time. The participants were initially directed to sit in the simulator and view the entertainment video displayed on the tablet PC. During this period, a black screen was displayed on the main display. TO alerts were triggered with beep sounds at random intervals between 60–120 s into the viewing. The participants were instructed to gather information from the driving scenes displayed on the monitor within a predetermined lead time. After the video ended, the participants were asked to respond to the SAGAT, SART, and perceived safety questions. In the practice session, the participants were trained with a 5-second video, and a 3- or 10-second video was used in the main session. The procedure of Session 2 was similar to that of Session 1 but without a predetermined lead time for TO. Rather, the participants were instructed to press a button to end the driving scene when they believed they had acquired all necessary information for driving. The stimulation was turned off after 30 s if a participant did not complete the video. The reason why the maximum length of video was set 30 s is because the average takeover reaction time when people have 60 s of time budget is around 20 s (Tan & Zhang, 2022).

2.4. Measures

Data on drivers' SA and subjective responses regarding safety, response time, and eye-tracking metrics were collected (Table 2). The SA scores were measured using the SAGAT and SART questionnaires. The SAGAT was developed to assess participants' objective SA levels and comprised 12 items. It was designed based on perception, comprehension, and prediction levels, comprising four items

Table 2

Dependent measures used in the study.

	Measure	Dimension	Unit	Description
Situation Awareness	SAGAT (Objective)	L1 (Perception)	%	The correct response rate on the environment during the given lead time.
		L2 (Comprehension)		
		L3 (Projection)		
	SART (Subjective)	Overall Demand	7 points Likert scale	The degree of attentional demand, attentional supply, and understanding of the situation.
		Supply		
		Understanding Overall		
Subjective Measure	Safety/Risk Perception		7 points Likert scale	The perceived safety of the situation is to be taken over when the video ends.
Response time			sec	Time considered sufficient to gain enough SA to regain control of the vehicle
Eye-Tracking Metric	Gaze behavior	% glance in the AOI	%	The percentage of glances during the lead time given in each of the 4 AOI

each. The items were designed as multiple-choice questions and included an “I’m not sure” choice. SAGAT method has its limitation in that as it utilizes freeze probe techniques, the results may rely on the memory capacity rather than situation awareness itself (Salmon et al., 2009). To overcome the limitation, only information for two-seconds before the video ended was considered as answer. The reason for setting the threshold of two-seconds was from the prior knowledge that distraction over two seconds significantly increase the crash rate (Klauer et al., 2006), which consequently generated a threshold of two-seconds rule such as AttenD (Kircher & Ahlström, 2009). The SART measures participants’ subjective SA levels, comprising 10 items under three dimensions: attentional demand, attentional supply, and understanding. Responses were recorded on a 7-point Likert scale. The final score was calculated by subtracting the attentional demand score from the sum of the attentional supply and understanding scores (Bolton et al., 2021). SART is a subjective assessment method that is easy to use and applicable in various fields, but it has limitations, including the reliability issues commonly found in subjective evaluation (Gawron, 2019). In addition, the perceived safety scores were obtained using a single-item questionnaire (“When the video ended, how safe did you perceive the situation to be for driving?”). Additionally, to assess participants’ level of immersion in non-driving-related tasks under each condition, we asked a single-item question with 7-points Likert scale (“How immersed were you in the entertainment video before the takeover (TO) alert was issued?”).

For gaze behavior, data on the proportion of glances directed toward four area of interest (AOI) during the TO period were collected. Previous research has divided the driver’s area of interest into four to five sections, including the road (center) section, human-machine interface (bottom) section, and left, right, and rear-view mirror sections (Lu et al., 2017; Goncalves et al., 2022). While some studies utilize a virtual mirror to display the passenger-side mirror in a single monitor condition (e.g., Lu et al., 2017), others exclude it to enhance realism (Li et al., 2022; Chandrasekaran et al., 2019). Therefore, this study included the front window, instrument cluster, left window, and rear-view mirror (Fig. 2), which have been identified as crucial for gathering driving-related information for SA development. The right-wing mirror section was not included in this study. In addition, during the second session, the participants were asked to indicate when they felt they had fully gathered the necessary information for re-engaging manual driving. This study focused on SA rather than TO performance; therefore, TO measures were not collected.

2.5. Apparatus and experimental setting

The experiment was performed using a fixed driving simulator (Fig. 3). The experimental stimuli were provided through a 27-inch display with a 1920×1080 -pixel resolution, showing the frontal view and left-side window of a left-hand-drive vehicle (Fig. 2). The stimulus videos were designed using the City Car Driving Simulator. All TO scenarios were recorded from the driver’s perspective in a sedan. On the right side, a tablet personal computer (PC) with an 11-inch display was used to present a video to elicit immersion in the NDRT during autonomous driving. A Logitech Momo steering wheel and pedal system were installed to simulate the driving situation. Tobii Pro Glasses 3 were used to collect eye gaze behavior data.

The driving video was extracted from the City Car Driving simulation, featuring a 30-minute drive in both urban and motorway settings with traffic densities of 30 % and 100 %, respectively. The extraction conditions were as follows: in the first session, stimuli were required to elicit consistent responses on the SAGAT questionnaire for more than 2 s at the end of each stimulus. To ensure this, three primary stimuli were developed for each condition, and two researchers evaluated each one. The segment that received the highest agreement among the three stimuli was selected as the final stimulus. In the second session, it was stipulated that the vehicle must not stop for 30 s (the maximum response time), and the driving environment (urban or motorway) must remain unchanged throughout the duration.



Fig. 2. Description of area of interest (AOI). (1) Front window, (2) Instrument cluster, (3) Left window (including left wing mirror), and (4) rear-view mirror. Other areas were considered as out of the AOI.

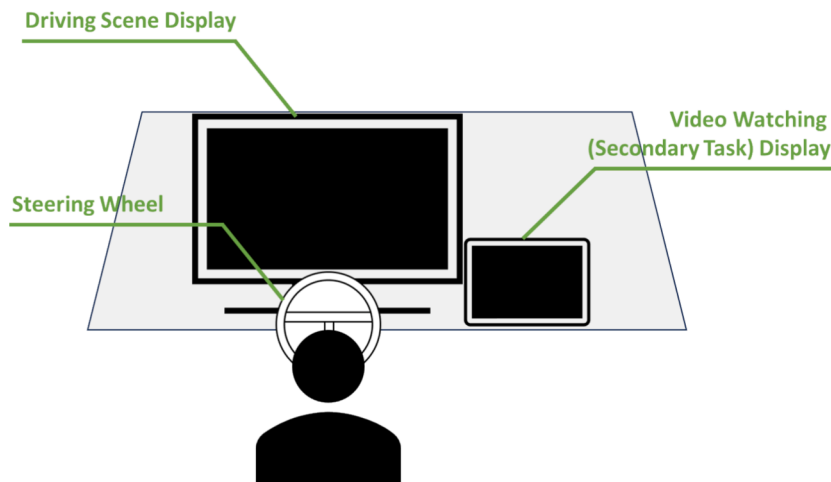


Fig. 3. Graphical representation of experimental settings.

2.6. Statistical analysis

To assess the influence of environmental and time budget factors on drivers' SA, a three-way analysis of variance (ANOVA) was employed, considering traffic density, road type, and time as within-subject factors. To compare situation awareness between the predefined lead time condition and the free lead time condition, we used data from both session one and session two in the ANOVA analysis. A post-hoc analysis using Tukey's test was conducted to examine specific differences. In addition, a one-way ANOVA was performed to compare drivers' SA levels based on time budget factors, with post hoc analysis conducted using Tukey's test. Gaze behavior analysis was performed using two-way ANOVA to explore the impact of the driving environment within each lead-time condition. All statistical analyses were performed using the IBM SPSS 27.

3. Results

3.1. Effect of environment factor and time budget on drivers' SA and perceived safety

The analysis of immersion levels before the takeover revealed no significant differences across conditions ($F(11, 604) = 0.556, p = 0.864$), with an overall high level of immersion ($M = 5.95, SD = 1.12$) in the NDRT. Table 3 shows the results of the descriptive data analysis of the SAGAT, SART, and perceived safety scores. The highest SAGAT score was obtained for low-traffic, highway, and 10-second lead time conditions ($M = 0.81, SD = 0.134$), whereas the lowest SAGAT score was obtained for high-traffic, urban, and 3-second lead time conditions ($M = 0.37, SD = 0.159$). Similar results were obtained for the SART scores ($M = 29.46, SD = 10.369; M = 4.95, SD = 10.141$). For perceived safety, the highest score was obtained for low-traffic, highway, and 10-second lead time conditions ($M = 5.36, SD = 1.564$), whereas the lowest score was obtained for high-traffic, urban, and 3-second lead time conditions ($M = 3.18, SD = 1.668$) and highway ($M = 3.18, SD = 1.620$).

Table 3

Descriptive Analysis of SAGAT, SART, and Perceived Safety score.

Independent Variables			N	SAGAT score		SART score		Perceived Safety score	
Traffic density	Road Type	Time		Mean	SD	Mean	SD	Mean	SD
Low	Highway	3 s	39	0.56	0.165	18.28	11.848	4.46	1.652
		10 s	39	0.81	0.134	29.46	10.369	5.36	1.564
		Free	76	0.60	0.171	27.43	9.839	5.29	1.364
	Urban	3 s	39	0.50	0.195	16.82	11.166	4.59	1.773
		10 s	39	0.59	0.151	24.10	10.718	4.79	1.641
		Free	76	0.58	0.181	25.95	10.106	5.20	1.327
High	Highway	3 s	39	0.40	0.149	9.00	10.496	3.18	1.668
		10 s	39	0.54	0.138	20.77	8.907	4.28	1.555
		Free	76	0.47	0.150	21.25	10.246	4.70	1.488
	Urban	3 s	39	0.37	0.159	4.95	10.141	3.18	1.620
		10 s	39	0.63	0.122	19.85	10.150	4.10	1.861
		Free	76	0.49	0.193	19.92	11.130	4.57	1.603

SAGAT score: 0 ~ 1 (0 %~100 % correct answer), perceived safety: 7 points likert scale.

A three-way ANOVA was performed to examine the effects of traffic density, road type, and time budget factors on the SAGAT, SART, and perceived safety scores (Table 4). The results showed a significant interaction between the effects of traffic density, road type, and time on the SAGAT score ($F(2,604) = 10.557, p < 0.001$). The results showed a significant main effect of traffic density ($F(1,604) = 78.674, p < 0.001$), road type ($F(1,604) = 6.582, p = 0.011$), and time budget factor ($F(2,604) = 52.289, p < 0.001$) on the SAGAT and SART scores ($F(1,604) = 76.578, p < 0.001, F(1,604) = 7.621, p = 0.006, F(2,604) = 68.980, p < 0.001$). For the perceived safety score, the results showed a significant main effect of traffic density ($F(1,604) = 51.250, p < 0.001$) and time budget factor ($F(2,604) = 24.900, p < 0.001$).

The results of the interaction effect showed only statistically significant two-way interaction between traffic and road type ($F(1,604) = 21.922, p < 0.001$) and a three-way interaction effect for SAGAT score ($F(2,604) = 10.557, p < 0.001$). We conducted a two-way interaction examining the interaction between road type and traffic density at the different levels of time factor. For 10 s condition, the results suggest that there is a significant interaction between road type and traffic ($F(1,152) = 52.198, p < 0.001$). During the 10-second condition, high traffic decreased SAGAT score significantly for highways compared to urban areas. While for 3 s and free condition, there was no significant difference in SAGAT score between road type across traffic density ($F(1,152) = 0.457, p = 0.500$ and $F(1,300) = 0.868, p = 0.352$).

The one-way ANOVA results showed a significant effect of time constraints on SAGAT, SART, and perceived safety. Fig. 4 shows the SAGAT, SART, and perceived safety scores according to lead time. The post-hoc analysis showed that SAGAT scores significantly differed between each condition, with the highest score for the 10-second condition ($M = 0.644$), followed by the free ($M = 0.534$) and 3-second ($M = 0.456$) conditions. The SART score for the 3-second condition ($M = 12.26$) significantly differed from those of the 10-second ($M = 23.54$) and free ($M = 23.64$) conditions (Fig. 4). Similarly, the perceived safety for the 3-second condition ($M = 3.85$) significantly differed from those of the 10-second ($M = 4.63$) and free ($M = 4.94$) conditions.

In addition, significant differences were observed in the subdimension for SAGAT (L1: perception, L2: comprehension, and L3: projection), as well as for SART (Demand, Supply, and Understanding; Table 5). The post-hoc analysis results of the SAGAT scores showed a significant difference between each time condition for SAGAT L1 (perception) and overall SAGAT score. For SAGAT L2 (comprehension), the results for the free and 3-second conditions significantly differed from those of the 10-second condition. By contrast, for SAGAT L3 (projection), the results for the 3-second condition significantly differed from those of the 10-second and free conditions. The SART scores for the 3-second condition significantly differed from those of the free and 10-second conditions for demand, supply, and understanding dimensions. The same was observed for their overall SART scores.

3.2. Response time based on traffic density and road type

Fig. 5 shows the distribution of drivers' response times across various driving scenarios. In all conditions, the response time distributions exhibited a right-skewed shape, indicating that most participants responded to TO early, with fewer participants responding later. The mean response time for all conditions was over 10 s. However, the distribution graph shows that the participants were predominantly concentrating before the 10-second mark. The median response time was less than 10 s, except for the high traffic in urban area conditions (D in Fig. 5). Low traffic on highways (A in Fig. 5), low traffic in urban areas (B in Fig. 5), and high traffic on highways (C in Fig. 5) exhibited similar distribution shapes, whereas the distribution for high traffic in urban areas was less concentrated. In addition, the response time ranged from 2.4 to 30 s. These results indicate that individual differences significantly affect the response time. Some participants were not ready to drive 20 s after the TOR, particularly in high-traffic and urban areas.

The mean response time increased when the traffic density was high under both highway and urban road conditions (Table 6). The average response times for highways were 10.65 s ($SD = 6.277$) and 11.47 s ($SD = 6.868$) for low and high traffic densities, respectively. For urban roads, the mean response times were 10.04 s ($SD = 5.748$) and 13.35 s ($SD = 7.805$) for low- and high-density conditions. The two-way ANOVA demonstrated no significant interaction between traffic density and road type with respect to response time ($F(1,300) = 2.626, p = 0.106$). Examining the main effects on response time, a significant impact was observed for traffic density ($F(1,300) = 7.142, p = 0.008$), whereas no significant main effect was observed for road type ($F(1,300) = 0.686, p = 0.408$). These results indicate that the participants required more time to prepare for driving, particularly under high-traffic conditions.

Table 4
Results of three-way ANOVA on SAGAT, SART, Perceived Safety.

Independent Variable	Significance level (<i>p</i> -value)		
	SAGAT	SART	Perceived Safety
Traffic	0<.001	0<.001	0<.001
Road type	0.011	0.006	0.291
Time	0<.001	0<.001	0<.001
Traffic*Road type	0<.001	0.705	0.785
Traffic*Time	0.553	0.079	0.059
Road*Time	0.093	0.648	0.463
Traffic*Road type*Time	0<.000	0.323	0.728

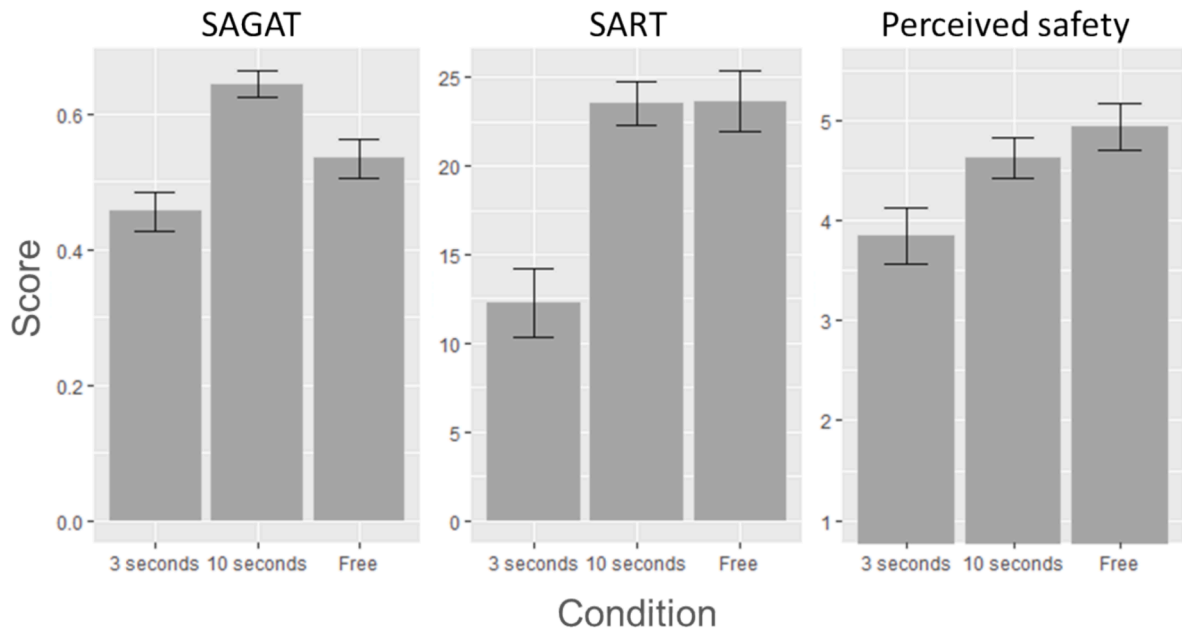


Fig. 4. Bar graph of SAGAT, SART, and Perceived Safety across the Lead Time.

Table 5

Descriptive Statistics and ANOVA Results for the Subdimensions of the SAGAT and SART Questionnaires.

Measure	Dimension	Descriptive Results				ANOVA			
		Free Mean [Min-Max] (SD)		Short (3 sec) Mean (SD)	[Min-Max]	Long (10 sec) Mean (SD)	[Min-Max]	<i>F</i> (2, 613)	Sig.
SAGAT	L1(Perception)	0.60 (0.239)	[0–1]	0.45 (0.242)	[0–1]	0.71 (0.236)	[0–1]	44.89	0.000
	L2(Comprehension)	0.41 (0.256)	[0–1]	0.41 (0.277)	[0–1]	0.59 (0.273)	[0–1]	25.16	0.000
	L3(Projection)	0.59 (0.318)	[0–1]	0.51 (0.315)	[0–1]	0.64 (0.268)	[0–1]	7.578	0.001
	Overall	0.53 (0.182)	[0–0.917]	0.46 (0.183)	[0–0.833]	0.64 (0.170)	[0.167–1]	43.667	0.000
SART	Demand	10.24 (4.046)	3–28	11.79 (4.475)	3–21	9.79 (3.95)	3–21	10.569	0.000
	Supply	19.16 (4.655)	4–28	13.51 (5.338)	4–28	18.81 (4.598)	5–28	76.785	0.000
	Understand	14.72 (3.512)	5–21	10.54 (3.881)	3–21	14.52 (3.588)	4–21	74.739	0.000
	Overall	23.64 (10.758)	[–10 – 46]	12.26 (12.152)	[–14 – 46]	23.54 (10.654)	[–3 – 46]	60.816	0.000

3.3. Drivers' eye gaze behavior based on environmental and time budget factors

Fig. 6 shows the gaze proportions for each AOI over time. In the 3-second condition (first column), a substantial portion of the gaze was directed toward the front window, particularly when the TOR was issued. After 1–1.5 s following the TOR, the participants shifted their visual attention to other AOIs. These findings suggest that drivers may face challenges in perceiving their driving status or potential hazards on the side and rear during the early stages of TO. Furthermore, the analysis revealed that the front window area varied with environmental factors. The two-way ANOVA results indicated a significant difference in the gaze ratio on the front window across all conditions. Specifically, in the 3-second condition, traffic density significantly influenced the gaze ratio on the front window ($p = 0.011$), indicating that the participants directed their gaze more toward the front window during high-traffic-density situations. In addition, a statistical difference in the gaze ratio on the rear-view mirror was observed in the 3-second condition. The participants gazed at the rear-view mirror less when traffic density increased ($p = 0.006$).

In the context of the 10-second condition, the gaze ratio on the front window increased at the initiation of the TO. However, the gaze ratio for each AOI stabilized approximately 3 s after the TOR was issued. Notable differences in the gaze ratio on the front window were observed in the 10-second condition. Under this condition, both traffic density ($p < 0.001$) and road type ($p = 0.006$) significantly

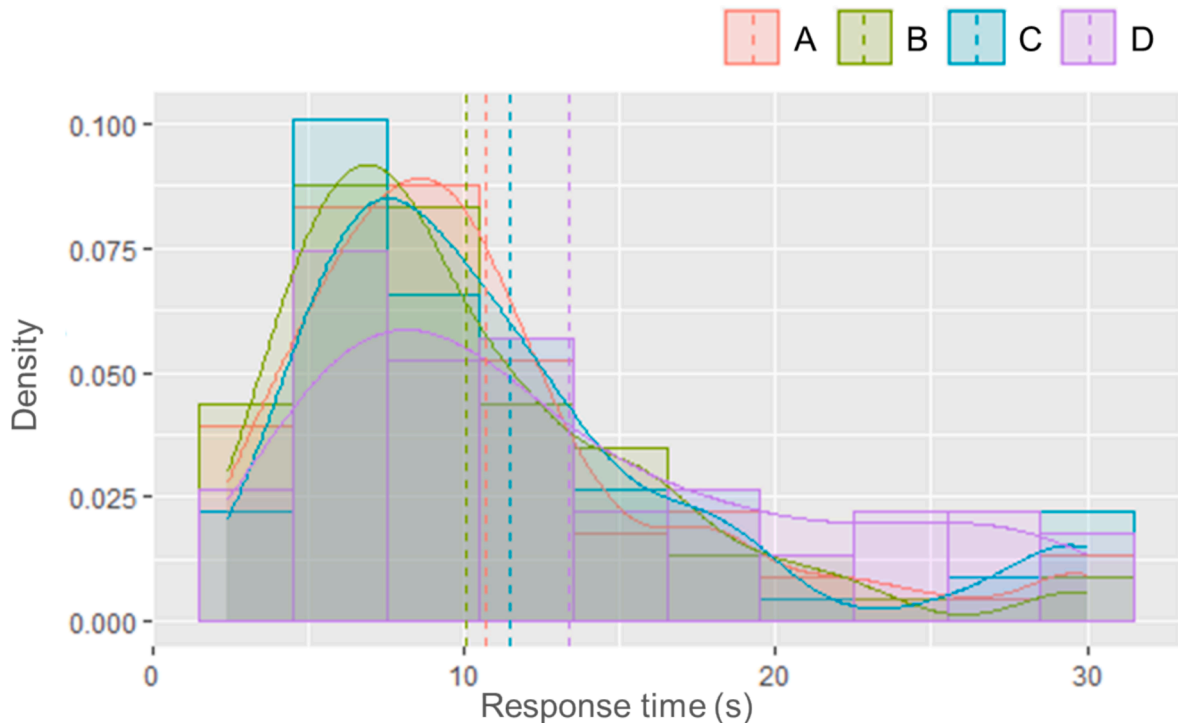


Fig. 5. Histogram and density distribution of response time across conditions. (A) Low Traffic in the Highway, (B) Low Traffic in Urban Areas, (C) High Traffic in the Highway, and (D) High Traffic in Urban Areas.

Table 6
Descriptive Statistics of Response Time across Experiment Conditions.

Traffic density	Road type	Mean	SD	Median	[Min-Max]
Low	Highway	10.65	6.277	9.38	2.42–30
	Urban	10.04	5.748	8.33	2.40–30
High	Highway	11.47	6.868	9.52	3.20–30
	Urban	13.35	7.805	11.13	2.78–30

influenced the gaze ratio directed toward the front window. Participants tended to focus more on the front window in high-traffic-density situations and when driving in urban areas. In addition, a statistical difference in the gaze ratio on the instrument cluster was evident in the 10-second condition, with the participants directing less attention toward the instrument cluster in high-traffic situations ($p < 0.001$).

The third column illustrates the gaze ratio under free lead time conditions, where the participants could halt the simulation once they believed they had gathered sufficient information for driving. Therefore, a decline in the graph over time indicated that some participants were prepared to start driving, leading to an overall reduction in the percentage. Notably, a distinct trend was observed in urban environments with high traffic density, with a gradual decrease in the slope compared with other conditions. This indicated that the participants in these high-demand settings tend to require a longer information-gathering period before initiating driving. A statistical difference in the gaze ratio on the front window was observed under the free lead time condition. Specifically, only the road type significantly influenced the gaze ratio directed toward the front window ($p = 0.012$). The participants demonstrated a higher gaze ratio in the front window while navigating urban areas under free lead time conditions.

4. Discussion

4.1. Effects of environmental and time budget factors on SA

Our first research question was ‘environmental factors and time budget factors influence drivers’ SA and perceived safety?’ The results of this study revealed the significant effect of traffic density, road type, and time on the SA scores and perceived safety. The more complex the situation, the lower the SA and perceived safety scores. The SAGAT score measures drivers’ SA based on the correct answers regarding the driving situation, which is linked to the amount of information the drivers can retain and use to proceed with the driving task. The SART and perceived safety scores are self-reported measures of how confident and comfortable they feel in the

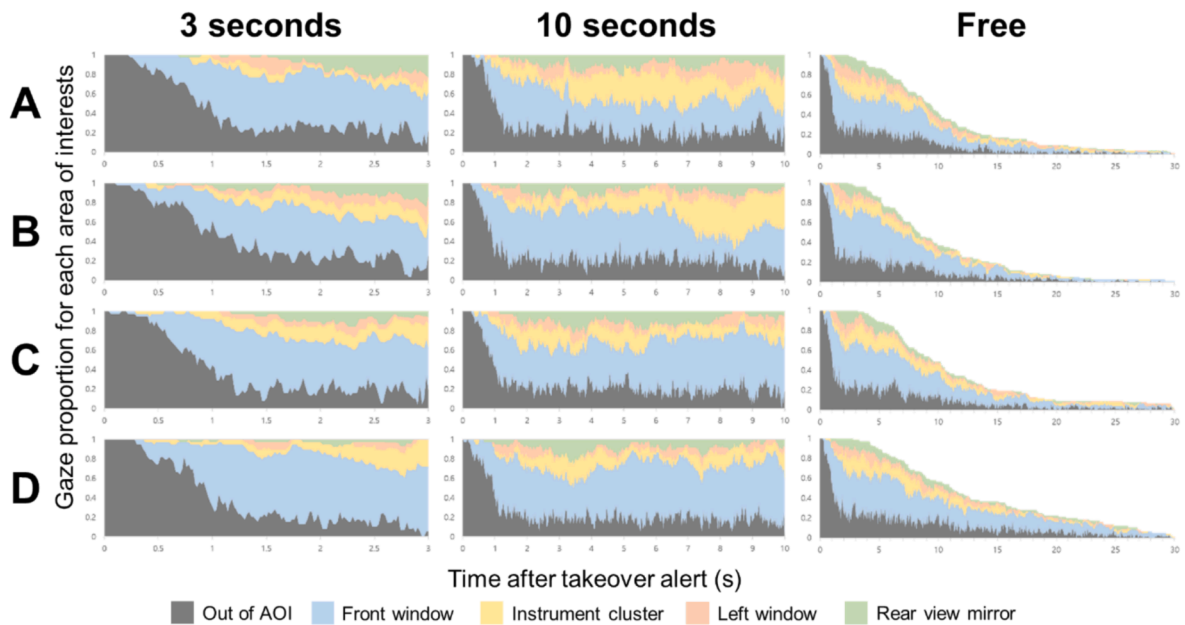


Fig. 6. Gaze Proportion on Areas of Interest (AOIs) under Varied Experimental Conditions. (A) Low Traffic in the Highway, (B) Low Traffic in Urban Areas, (C) High Traffic in the Highway, and (D) High Traffic in Urban Areas.

situation. However, a significant main effect was observed in both cases, featuring higher scores in less complex situations with longer times and lower scores in complex situations with shorter times.

Particularly, for the SAGAT score, the results suggest a significant two-way interaction effect on traffic density and road type. Especially, the high traffic density significantly decreased the SAGAT score during highway compared to urban areas. Likely, we can argue that for measuring the SAGAT score urban areas might provide more complex visual information. With regard to the 3-way interaction effect, the interaction effect between traffic density and road type differs across time factors. That is, it only showed statistically significant interaction during the 10 s conditions, where the participants had a longer time to increase their SA level. In this extended time condition, the negative impact of traffic density on SAGAT scores was notably reduced in urban areas. This indicates that drivers maintained consistent situation awareness levels regardless of surrounding vehicle density, provided they had adequate time for preparation but no immediate takeover option. This finding can be attributed to the inherent characteristics of urban environments, where lower speeds create less variability in the driving situation. The reduced speed in urban areas naturally provides a more stable and predictable context for takeover preparation, allowing drivers to maintain their situation awareness more effectively, even as traffic density increases.

In terms of response time and SA level, the results demonstrated that complex driving scenarios required longer response times to develop SA. However, these results differed from those of previous studies. For instance, [Jia & Du \(2023\)](#) demonstrated a shorter TO response time and lower SA score for high-traffic-density scenarios. This discrepancy can be attributed to the difference in the experimental setting and the task given to the drivers. This study primarily focused on SA and excluded the influence of time pressure on the TO task; we aimed to avoid the influence of time pressure posed by the TO task and the limitation of the experimental setting involving participants expecting a TO task that might influence a faster response, thereby influencing SA. Therefore, although complex driving situations required longer response times, the SA scores were lower for these situations.

A noteworthy result of this study was the participants' self-response regarding the required amount of time for drivers to gain sufficient information for driving. Previous studies revealed the positive impact of lead time on SA ([Wu et al., 2023](#)). In this study, a higher level of SA was expected when the participants were given free time to respond and prepare compared with the lead-time trials. However, as shown in [Fig. 4](#), the SAGAT score was significantly higher in the 10-second condition than in the free condition, and no significant difference was observed in the SART score between the two conditions. These results are consistent with those of [Tan & Zhang \(2021\)](#), in which the SA score was not linearly correlated with the lead time owing to the presence of a convergence point. [Table 6](#) indicates that the participants believed that they had gained sufficient SA to assume vehicle control near the 10-second time interval.

4.2. Driver's response time according to environmental factors

Our second research question consisted of 'How does the response time to indicate sufficient SA differ based on traffic density and road type?' An analysis of the response time distribution revealed that drivers demonstrate different levels of readiness to drive, influenced by environmental factors. In particular, the participants exhibited slower responses to the TOR in scenarios requiring

substantial information gathering, such as in high-traffic urban areas. This observation aligns with the logical expectation that longer lead times, known to enhance SA (Wu et al., 2023), prompt drivers to seek additional information when faced with a higher cognitive load. In situations requiring the processing of abundant information, drivers tended to prioritize the acquisition of relevant data, resulting in slower responses to the TOR. Notably, the response time distribution was right-skewed, indicating a prevailing trend toward faster responses. However, a notable density was observed on the right side, suggesting that some participants were not ready to drive even 20 s after the TOR issuance. Notably, the proportion of participants exhibiting delayed responses was high, particularly in high-traffic urban areas. This finding suggests that although the average drivers can attain an appropriate SA level within the lead time range suggested in previous studies (12–16 s – Lu et al., 2017; Tan & Zhang, 2022), some individuals encounter challenges in successfully assuming vehicle control.

4.3. Effects of environmental and time budget factors on Drivers' gaze patterns While redeveloping SA

Our last research question was 'How do environmental and time budget factors affect drivers' gaze behavior when redeveloping SA?' The gaze behavior under various conditions was analyzed. Consistent with the findings of Lu et al. (2017), the participants tended to focus on the front window when presented with a driving situation and subsequently redirected their gaze to other AOIs. Moreover, the findings of this study support the notion that the participants were predominantly fixated on the front window compared with other AOIs, such as the rear-view and wing mirrors (Gonçalves et al., 2022; Lu et al., 2017). In addition, participants' gaze behavior revealed that drivers, to some extent, were fixated on the instrument cluster when the TOR was issued. This indicated that drivers anticipated relevant information regarding the TO situation from the instrument cluster and directed their attention accordingly.

Furthermore, variations in the gaze behavior influenced by environmental settings and lead time conditions were observed. Specifically, under short lead time conditions (3 s), traffic density significantly affected gaze behavior, with high traffic leading to increased fixation on the front window. With a longer lead time (10 s), traffic density and road type influenced gaze behavior, highlighting the tendency for individuals to focus on the front window when navigating urban areas with high traffic density. Notably, an increased allocation of visual attention to the front window when driving in urban areas was observed in scenarios where participants had control over the TO (free lead time). This observation suggests a potential vulnerability to events occurring in other AOIs, emphasizing the significance of understanding gaze behavior under such conditions.

4.4. Limitation

The generalizability of the findings of this study should be approached by considering their inherent limitations. One of the limitations of this study is the methodological constraints associated with the use of SAGAT and SART for assessing situation awareness (SA). The SAGAT, by its nature, has the potential to alter participants' gaze behavior, as it interrupts the task to query the participant, which can influence their subsequent attention patterns. Similarly, the SART may induce learning effects, where repeated exposure to the tool leads to improvements unrelated to the participant's actual situational awareness, thereby skewing the results. These factors represent significant challenges in the accurate measurement of SA and should be considered when interpreting the findings of studies utilizing these instruments.

Using a fixed-based driving simulation environment featuring a single monitor with a limited field of view restricted the participants from observing the right side of the driving scene, including the right-wing mirror and window. Consequently, the collected data only partially represents natural gaze behavior during TO scenarios. Another limitation is that this study examined two distinct road types (highways and urban areas), considering the divergent visual distractions each presents. However, this methodological choice limited control of the vehicle speed, potentially affecting the participants' response times. Furthermore, this study exclusively focused on the TO phase, neglecting the post-TO driving performance and driver reactions to potential hazards.

5. Conclusions

This study investigated the impacts of time constraints and environmental factors on driver SA development under HAD scenarios. The first research question examined the differences in SA levels based on traffic density, road type, and time constraints. The findings revealed that traffic density and road type significantly influenced the SA, with a lower SA observed when vehicle control was regained in high-traffic and urban areas. Moreover, time constraints proved significant, indicating that longer lead times were associated with higher SA. The second research question explored the variations in SA levels when drivers were given sufficient lead time to resume control. While the SART scores remained comparable across conditions with free and 10-second lead times, the SAGAT scores were lower than those in the 10-second condition. This suggests that, although drivers may have a lower objective SA level when taking over freely, they exhibit confidence in their TO abilities, potentially indicating a reduced focus on information gathering on the driving scene when not constrained by limited TO time. The final research question focused on drivers' gaze behavior with respect to different environmental and time budget factors. The results indicated that the drivers predominantly concentrated their visual gaze on information from the front window. Especially for low-traffic density scenarios, the proportion of visual gaze in other AOIs increased compared to high-traffic density scenarios. Overall, this study contributes to existing research by highlighting that the SA development process can differ based on environmental factors and time constraints. This underscores the importance of providing TOR strategies based on specific TO scenarios. Moreover, the findings of this study have implications for designing effective interventions and strategies to enhance drivers' readiness during transitions from automated to manual driving.

Nonetheless, future studies should address the identified study limitations by investigating the impact of a more realistic and

immersive driving simulation environment that addresses the limitations of a fixed-based setup, exploring a broader spectrum of road types beyond highways and urban areas to capture a more diverse range of driving scenarios, and investigating the influence of environmental factors on the driving performance following TO events.

CRediT authorship contribution statement

Young Woo Kim: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Sol Hee Yoon:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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